INTRODUCTION

Life of a cell is entirely based on planar sandwiches called membranes that are capable of protecting and separating the cells from the world around. These membranes, generally, are impermeable to macromolecules but can facilitate movement of certain molecules across it and contribute to maintain homeostasis at a greater level .

Membranes are comprised of two layers of lipids with embedded proteins. They act as two-dimensional entity, with a polar group, facing water exterior and hydrocarbon tails facing the interior side of cell. While membranes encircling the entire cell act as barriers against external environment, membrane around each organelle creates unique interior space for specialized biochemical reactions of that organelle, finally, contributing to the life of a cell.

The current knowledge on membrane has its origin from the findings of E.coli in the year 1855 which described that a membrane consists of lipid bilayer. In the following century, biochemical experts and X-ray experts had postulated the membrane as a surface coating with proteins diffused in the place of membrane or floating or spanning, with lipid anchors.

The following content of the chapter will provide you a detailed knowledge on structure and composition of a membrane.

1.1. CHEMICAL COMPOSITION

1.1.1. Lipid

It is understood from the introductory notes that lipids can form the framework of membrane, with the anchored proteins. The lipid molecules are less than 1000 Dalton in size and consist of aliphatic/aromatic hydrocarbons. There are various types of lipids that form the bilayer structure of a membrane.

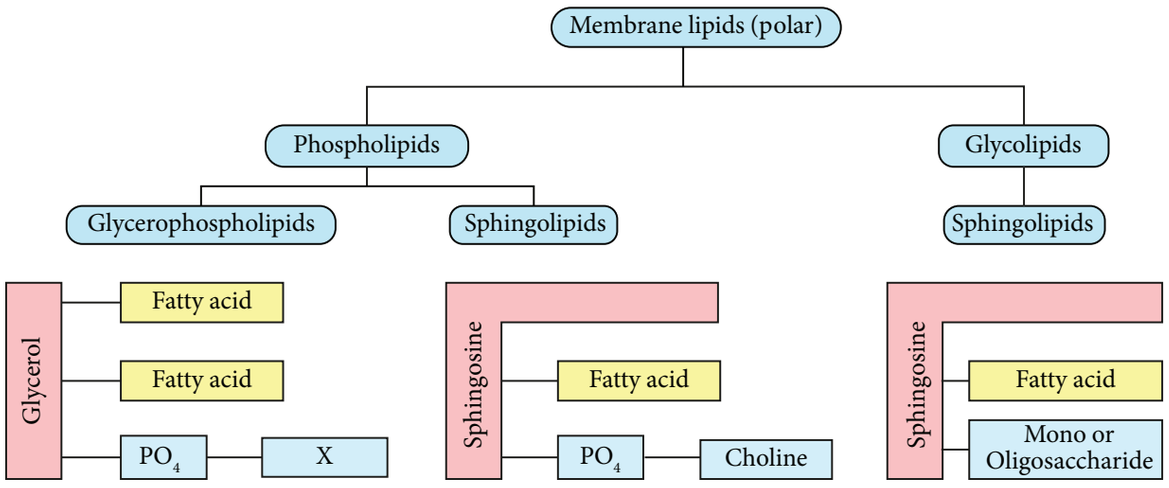

Figure 1.1 Lipid framework of a bilayered membrane

XII U1 Cell Membrane.indd 2 4/8/2020 3:03:27 PM